

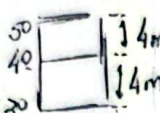
Componente Curricular: Física - Dinâmica

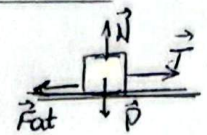
Professor(a): Juliana C. Padua

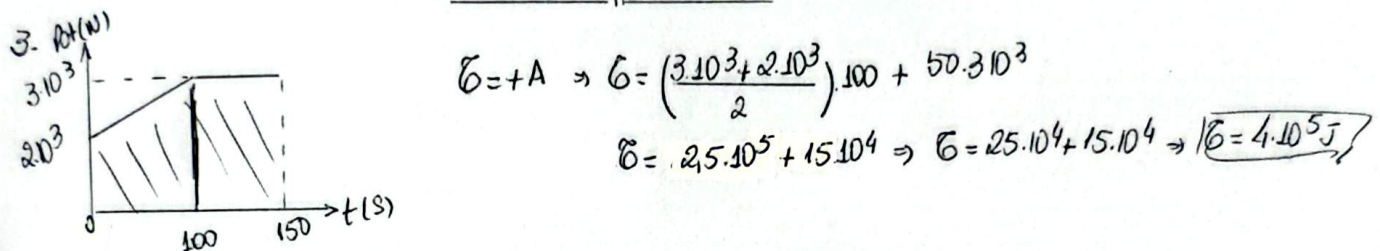
Conteúdo: Potência - Exercícios Resolvidos

Série: 2ª

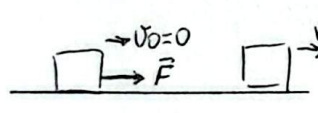
Data: Abril - 2026

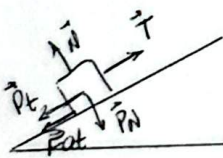
1.  * $F = P$ (MRU) $Pot = \frac{G}{\Delta t} \Rightarrow Pot = \frac{F \cdot d}{\Delta t} \Rightarrow Pot = \frac{600 \cdot 8}{10} \Rightarrow Pot = 480 W$
 $F = 600 N$

2. $v_{rel} (v = 0,5 m/s)$ * $F_{at} = \mu \cdot N$ *  * $T = F_{at}$ * $Pot_u = T \cdot v$
 $F_{at} = 0,1 \cdot 5000$ $T = 500 N$ $Pot_u = 500 \cdot 0,5 \Rightarrow Pot_u = 250 W$
 $F_{at} = 500 N$



4. * M_1  * $MU (F_R = 0)$ * $G_T = T \cdot h$ * $Pot_t = \frac{G}{\Delta t} \Rightarrow Pot_t = \frac{200}{10} \Rightarrow Pot_t = 20 W$
 $T = P \Rightarrow T = 10 N$ $G_T = 10 \cdot 20$ $G_T = 200 J$ $Opca: v = \frac{\Delta S}{\Delta t} \Rightarrow v = \frac{20}{10} = 2 m/s$ * $Pot = T \cdot v$
 $Pot = 10 \cdot 2 = 20 W$

* M_2  * $v_0 = 0$ * $v = 10 m/s$ * $v = v_0 + at$ * $F_R = m \cdot a$ * $\Delta S = \frac{a \cdot t^2}{2}$ * $G_F = F \cdot d$
 $t = 2 s$ $10 = a \cdot 2$ $F = 3,5$ $\Delta S = \frac{5 \cdot 2^2}{2} = 10 m$ $G_F = 15 \cdot 10 \Rightarrow G_F = 150 J$
 $a = 5 m/s^2$ $F = 15 N$ * $Pot = \frac{G}{\Delta t} \Rightarrow Pot = \frac{150}{2} \Rightarrow Pot = 75 W$

5.  $\Delta S = 2 m$ * $v_{rel} \Rightarrow F_R = 0$ * $P_H = P \cdot \cos \alpha$ * $P_t = P \cdot \sin \alpha$ * $F_{at} = \mu \cdot N$ * $T = P_t + F_{at}$
 $P_H = 100 \cdot 0,8 = 80 N$ $P_t = 100 \cdot 0,6 = 60 N$ $F_{at} = 0,5 \cdot 80 = 40 N$ $T = 60 + 40 = 100 N$

* $Pot = \frac{G}{\Delta t} \Rightarrow Pot = \frac{T \cdot d}{\Delta t} \Rightarrow Pot = \frac{100 \cdot 2}{10} \Rightarrow Pot = 20 W$

Componente Curricular: Física - Dinâmica

Professor(a): *Rafaela C. Rêdua*

Série: 2ª

Data: *16/01/2025*

Conteúdo: *Pendentes - Exercícios - Resolvidos*

1. Motor recebe $\Rightarrow P_{otl} = 200W$
Realiza um $\bar{C} \Rightarrow P_{otn} = 180W$

a) $P_{otl} = P_{otu} + P_{otd}$
 $200 = 180 + P_{otd}$
 $P_{otd} = 20W$

b) $\eta = \frac{P_{otu}}{P_{otl}} \Rightarrow \eta = \frac{180}{200} = 0,9$ ($\eta = 90\%$)

2. Motor recebe $\Rightarrow P_{otl} = 200W$
Realiza $\bar{C} \Rightarrow \bar{C}_u = 8640J$
 $\Delta t = 1 \text{ min} = 60s$

* $P_{otu} = \frac{\bar{C}_u}{\Delta t}$
 $P_{otu} = \frac{8640}{60} \Rightarrow P_{otu} = 144W$

* $\eta = \frac{P_{otu}}{P_{otl}} \Rightarrow \eta = \frac{144}{200} = 0,72$
($\eta = 72\%$)

3. * $P_{otu} = F \cdot v$
 $P_{otu} = 1000 \cdot 30$
 $P_{otu} = 3 \cdot 10^4 W$

* $\eta = \frac{P_{otu}}{P_{otl}} \Rightarrow 0,2 = \frac{3 \cdot 10^4}{P_{otl}} \Rightarrow P_{otl} = \frac{3 \cdot 10^4}{2 \cdot 10^{-1}} = 1,5 \cdot 10^5 W$ ou $P_{otl} = 15 \cdot 10^4 W$

4.  $v_{ck} \Rightarrow F_R = 0$
* $T = P$
 $T = 200N$

* $P_{otu} = T \cdot v$
 $P_{otu} = 200 \cdot 2$
 $P_{otu} = 400W$

* $\eta = \frac{P_{otu}}{P_{otl}} \Rightarrow 0,8 = \frac{400}{P_{otl}}$
 $P_{otl} = 500W$

* $P_{otl} = P_{otu} + P_{otd}$
 $500 = 400 + P_{otd}$
 $P_{otd} = 100W$

5. * $P_{otl} = 100 \text{ HP}$
 $P_{otl} = 100 \cdot 0,75 \text{ kW}$
 $P_{otl} = 75 \text{ kW}$

* $\bar{F}_R = m \cdot a$
 $F = 1000 \cdot 3$
 $F = 3 \text{ kN}$

* $\bar{C} = F \cdot d$
 $\bar{C} = 3 \cdot 10^3 \cdot 2 \cdot 10^2$
 $\bar{C} = 6 \cdot 10^5 J$

* $P_{otu} = \frac{\bar{C}}{\Delta t}$
 $P_{otu} = \frac{6 \cdot 10^5}{10}$
 $P_{otu} = 6 \cdot 10^4 W$

$\eta = \frac{P_{otu}}{P_{otl}}$
 $\eta = \frac{6 \cdot 10^4}{75 \cdot 10^3} \Rightarrow \eta = 0,8$
($\eta = 80\%$)